

MYP4 Sciences - Criterion A Formative Assessment (Units 1-5)

(90 minutes, printed version)

Name: _____

Class: _____

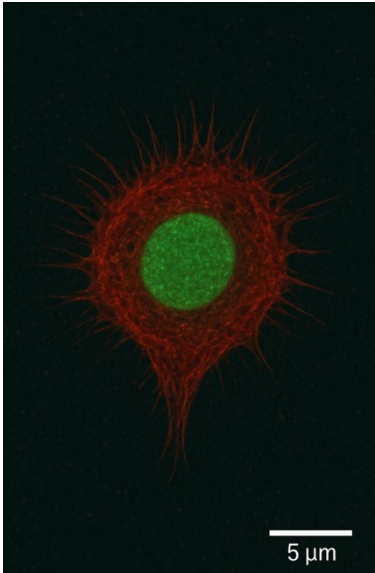
Date: _____

- $F_g = G \frac{m_1 m_2}{r^2}$
 - $Q = mc\Delta T$
 - Avogadro's constant: $6.022 \times 10^{23} \text{ mol}^{-1}$
- Radius of Earth = 6'371 km

Section A: Knowledge and Understanding

1. **State** the SI unit for mass and the prefix for 10^{-6} . [1-2]
2. **Order** from largest to smallest: Hydrogen atom, Mitochondrion, Carbon nucleus, Glucose molecule, Human egg cell. [3-4]
3. **Describe** the purpose of a control group in an experiment. [3-4]

4. A student measures a cell image with a 5 μm scale bar. **Calculate** the cell's actual size (entire red segment). [5-6]



5. A scientist claims, "One experiment is enough to prove a theory." **Evaluate** this statement. [7-8]

6. **Draw** an atom and its subatomic particles [3-4]

Section B: Application and Analysis

7. **Identify** the subatomic particle(s) that:
a) Differ in isotopes. [1-2]

b) Are lost to form a Ca^{2+} ion. [1-2]

8. An element has 20 protons, 20 neutrons, and 18 electrons.
a) **State** its atomic number. [1-2]

b) **Explain** why it's charged. [3-4]

9. **Draw** a labeled diagram of a carbon-12 and carbon-14 atom, highlighting their differences. [5-6]

Carbon-12

Carbon 14

10. **Calculate** the gravitational force between Earth (5.97×10^{24} kg) and a 500 kg satellite 200 km above its surface. [7-8]

11. **Compare** ionic and covalent bonds in terms of electron behavior. [5-6]

12. **State** three trends in the periodic table, and their direction.

13. **Analyze** the trend in atomic radius across Period 3 (Na to Ar) on the periodic table. **Explain** how electron shielding and effective nuclear charge contribute to this trend. **[7-8]**

Section C: Evaluation and Synthesis

11. **Balance** the equation:
$$\text{Al} + \text{O}_2 \rightarrow \text{Al}_2\text{O}_3$$
 [3-4]
12. **Interpret** the electronegativity trend across Period 3. **[7-8]**

13. A 50 g sample of iron absorbs 450 J of heat, increasing its temperature by 10°C. **Calculate** its specific heat capacity. [5-6]



14. **Evaluate** whether a cold “bitterbal” crust means the filling is also cool, given:

- o Crust $c=300 \text{ J/kg/}^\circ\text{C}$
- o Filling $c=3500 \text{ J/kg/}^\circ\text{C}$

[7-8]

15. **Design** a method to estimate the number of fish in the lake in the image using sampling. [7-8]

A student reacts 12.6 g of magnesium (Mg) with excess hydrochloric acid (HCl). The balanced equation is:



16. **Calculate** the number of moles of Mg used.